CuraLM-70B: Adapting a Large Language Model to Clinical Text

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Abstract

CuraLM-70B adapts a Large Language Model to Medical QA through continued training on de-identified clinical notes and medical literature, followed by Instruction Tuning on medical question-answer pairs. The goal is to improve terminology, answer format, and clinical reasoning without relying on external retrieval. We evaluate on MedQA and PubMedQA. CuraLM-70B exceeds a general-purpose 70B model by about ten accuracy points on MedQA and improves on PubMedQA as well. However, rare conditions remain difficult, and the system is too large to run in hospitals without data-center hardware. The paper studies domain adaptation as a strong but resource-intensive path to clinical performance.

Keywords

Clinical adaptation, Instruction Tuning, Medical QA, MedQA

1. Introduction

Medical QA requires terminology and conventions that broad training may not cover. Retrieval-augmented generation [1] is an alternative, but CuraLM-70B instead adapts its internal representations. FactProbe [2] reports high hallucination rates for general systems, motivating domain-specific training. This paper measures how far adaptation can go before infrastructure becomes the limiting factor.

2. Method

CuraLM-70B begins with a 70-billion-parameter Large Language Model. Continued training uses de-identified clinical notes and medical literature filtered for duplication and identifiable content. Instruction Tuning then uses question-answer pairs with answer rationales removed from the input and concise final answers as targets. A curriculum moves from terminology normalization to multi-sentence clinical decisions. We retain separate validation splits for MedQA and PubMedQA, use temperature-free decoding, and report exact accuracy. A general-purpose 70B model is evaluated with the same prompts. The training recipe emphasizes domain vocabulary and instruction adherence rather than external tools, so the final system can be evaluated as a standalone Medical QA model.

3. Experiments

CuraLM-70B reaches 63.8% accuracy on MedQA versus 53.6% for a general-purpose 70B model. On PubMedQA it reaches 71.2% versus 64.9%. Error review finds that rare conditions and uncommon drug combinations remain fragile. The model requires data-center hardware and cannot be deployed directly on ordinary hospital workstations.

System	MedQA accuracy	PubMedQA accuracy
General-purpose 70B	53.6%	64.9%
CuraLM-70B	63.8%	71.2%

4. Conclusion

CuraLM-70B shows that domain adaptation and Instruction Tuning improve Medical QA, especially on MedQA. Yet rare conditions still trigger hallucinated details, and the 70B footprint makes deployment expensive. Future systems should combine domain expertise with lighter evidence mechanisms.

References

[1] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).

[2] Novak, H. (2022). FactProbe: A Benchmark for Measuring Hallucination in Large Language Models. Transactions on Language Evaluation (TLE).